

0 8 . 1

A non-competitive inhibitor decreases the rate of an enzyme-controlled reaction.

Explain how.

[3 marks]

0 8 . 2

A scientist investigated the hydrolysis of the protein casein.

The scientist:

- mixed a solution of a protease enzyme with a solution of casein
- then measured the casein concentration in the mixture at intervals
- controlled all relevant variables appropriately.

For this investigation, identify:

[2 marks]

the independent variable _____

the dependent variable _____

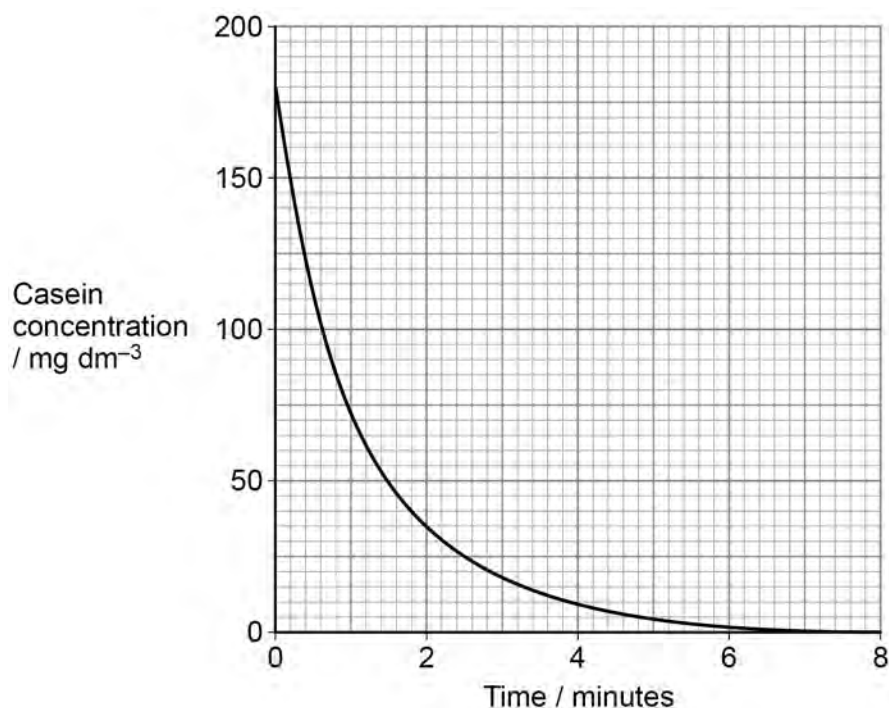
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0 8 . 3 Figure 10 shows the scientist's results.

Do not write
outside the
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Figure 10



Use **Figure 10** to determine the rate of casein hydrolysis at 2 minutes.

Show how you obtained your answer.

[2 marks]

Answer _____ mg dm⁻³ minute⁻¹

0 8 . 4 The scientist repeated the investigation but increased the temperature to the optimum temperature for this protease.

Sketch a line on **Figure 10** showing the results you predict if the investigation is repeated at the optimum temperature for the protease.

[2 marks]

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